Getting started with Cloud Volumes ONTAP in Google Cloud Platform

Cloud Manager 3.7

Ben Cammett
November 17, 2020

This PDF was generated from https://docs.netapp.com/us-en/occm37/task_getting_started_gcp.html on December 04, 2020. Always check docs.netapp.com for the latest.

Table of Contents

Getting started with Cloud Volumes ONTAP in Google Cloud Platform 1

Getting started with Cloud Volumes ONTAP in Google Cloud Platform

Get started with Cloud Volumes ONTAP by setting up GCP and then deploying Cloud Manager software from NetApp Cloud Central.

Cloud Manager must be installed in Google Cloud Platform in order to deploy Cloud Volumes ONTAP in GCP.

1 Set up your networking

Enable outbound internet access from the target VPC so Cloud Manager and Cloud Volumes ONTAP can contact several endpoints.

This step is important because Cloud Manager can't deploy Cloud Volumes ONTAP without outbound internet access. If you need to limit outbound connectivity, refer to the list of endpoints for Cloud Manager and Cloud Volumes ONTAP.

2 Set up GCP permissions and projects

Make sure that two sets of permissions are in place:

a. Ensure that the GCP user who deploys Cloud Manager from NetApp Cloud Central has the permissions in the Cloud Central policy for GCP.

   You can create a custom role using the YAML file and then attach it to the user. You'll need to use the gcloud command line to create the role.

b. Set up a service account that has the permissions that Cloud Manager needs to create and manage Cloud Volumes ONTAP systems in projects.

   You'll associate this service account with the Cloud Manager VM in step 6.

   - Create a role in GCP that includes the permissions defined in the Cloud Manager policy for GCP. Again, you'll need to use the gcloud command line.

     The permissions contained in this YAML file are different than the permissions in step 2a.

   - Create a GCP service account and apply the custom role that you just created.
   - If you want to deploy Cloud Volumes ONTAP in other projects, grant access by adding the service account with the Cloud Manager role to that project. You'll need to repeat this step for each project.

3 Set up GCP for data tiering

Two requirements must be met to tier cold data from Cloud Volumes ONTAP 9.7 to low-cost object storage (a Google Cloud Storage bucket):

a. Create a service account that has the predefined Storage Admin role and the Cloud Manager service account as a user.

   You'll need to select this service account later when you create a Cloud Volumes ONTAP working environment. This service account is different from the service account that you created in step 2.

b. Configure the Cloud Volumes ONTAP subnet for Private Google Access.

If you want to use data tiering with Cloud Volumes ONTAP 9.6, then follow these steps.

4 Enable Google Cloud APIs

Enable the following Google Cloud APIs in your project. These APIs are required to deploy Cloud Manager and Cloud Volumes ONTAP.

- Cloud Deployment Manager V2 API
- Cloud Resource Manager API
- Compute Engine API
- Stackdriver Logging API

5 Subscribe from the GCP Marketplace

Subscribe to Cloud Volumes ONTAP from the GCP Marketplace to ensure that there's no disruption of service after your free trial ends. You'll be charged from this subscription for every Cloud Volumes ONTAP PAYGO system that you create.

6 Launch Cloud Manager from NetApp Cloud Central

Cloud Manager software is required to deploy and manage Cloud Volumes ONTAP. It takes just a few minutes to launch a Cloud Manager instance in GCP from Cloud Central.

When you choose GCP as the cloud provider, you're prompted by Google to log in to your account and to grant permissions. Clicking "Allow" grants access to the compute APIs needed to deploy Cloud Manager.

7 Launch Cloud Volumes ONTAP using Cloud Manager

Once Cloud Manager is ready, just click Create, select the type of system that you would like to deploy, and complete the steps in the wizard. After 25 minutes, your first Cloud Volumes ONTAP system should be up and running.

Related links

- Evaluating
- Networking requirements for Cloud Manager
- Networking requirements for Cloud Volumes ONTAP in GCP
- Firewall rules for GCP
- What Cloud Manager does with GCP permissions
- Launching Cloud Volumes ONTAP in GCP
- Downloading and installing the Cloud Manager software on a Linux host

Copyright Information

Copyright © 2020 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.